

- 1** Solve for x.

$$x^2 + 5x + 6 = 0$$

- 2** Solve for x.

$$2x^2 + 7x - 4 = 0$$

- 3** Find all values of x for which the following holds.

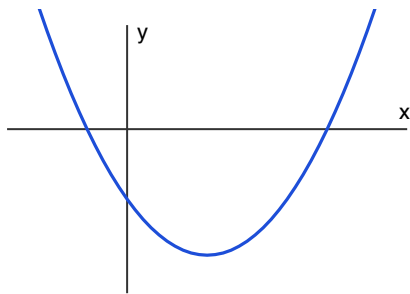
$$(x - 3)(x + 2) = 6$$

- 4** Solve, giving exact values.

$$3x^2 - 5x - 1 = 0$$

- 5 Find the x-intercepts and the turning point of the graph of

$$y = x^2 - 4x - 5$$



- 6 For which value of k does the graph of the following touch the x-axis exactly once?

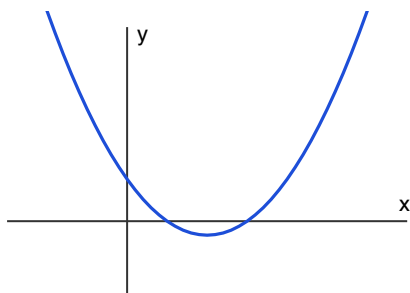
$$y = x^2 + 6x + k$$

- 7 Factorise fully.

$$\frac{1}{3}x^2 + 2x + \frac{8}{3}$$

- 8** The graph of the following is shown. Read off its x-intercepts and check them.

$$y = x^2 - 4x + 3$$



- 9(a)** Show that the following has no real solutions.

$$x^2 + 4x + 5 = 0$$

- 9(b)** Say what that means for the graph of $y = x^2 + 4x + 5$.